

# Roots of Evolution: An Immersive VR Journey Through Plant Evolution Using Digital Storytelling in Botany

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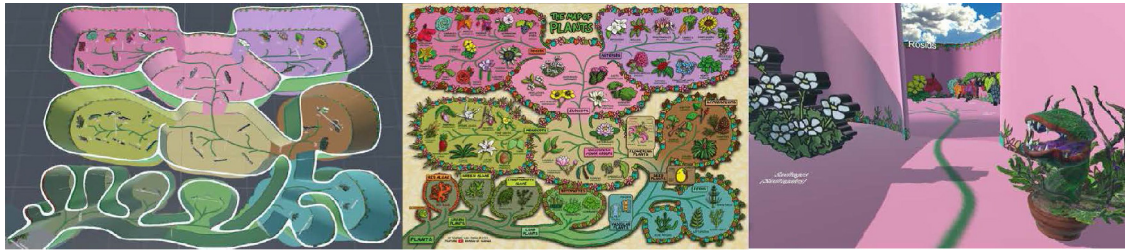


Figure 1: Overview of the 'Roots of Evolution' VR Museum, featuring the 3D layout, the 2D 'Map of Plants' by Dominic Walliman (the conceptual foundation for the exhibit), and an immersive corridor with interactive plant sections and an AI-driven guide.

## ABSTRACT

The public's growing awareness of sustainability and plant diversity makes teaching plant evolution a fascinating and emergent challenge for learners at every level. *Roots of Evolution* is an immersive Virtual Reality museum using digital storytelling to explore plant evolution interactively. Inspired by the "Map of Plants" by Dominic Walliman, this project aims to simplify complex concepts and foster curiosity through interactive multi-modal content, an intelligent AI agent, and a narrative-driven approach to 3D immersive media. Future expansions aim to include broader botanical topics and user testing to refine its educational impact, showcasing the potential of immersive technology and digital storytelling in science education.

**Index Terms:** Virtual Reality, Digital Storytelling, Education, Botany.

## 1 INTRODUCTION

The diversity of plant life on Earth results from half a billion years of evolution. Such an extended timeline and the diversity of life on Earth make this topic challenging to teach, especially since we are less aware of plants' diversity than animals' diversity [4]. The "Map of Plants" video from Dr. Walliman on his YouTube channel *Domains of Science*<sup>1</sup> offers an accessible and scientifically accurate representation of plant evolution. This video inspired the creation of *Roots of Evolution*, an immersive Virtual Reality (VR) museum for presenting the evolutionary journey of plants as an interactive experience through digital storytelling. The VR museum has several potential applications. It can be used as an educational tool in botany classes or as an exhibition feature in science museums to make this complex topic accessible to everyone. This extended abstract for IEEE VR presents the design, development, and potential impact of *Roots of Evolution*, showcasing how immersive technologies can transform how we teach and learn about the natural world.

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<sup>1</sup><https://www.youtube.com/watch?v=ONVpFtiD-fo>

## 2 BACKGROUND

VR has shown the potential to transform education by offering experiences that enhance engagement via immersion, interaction, imagination, and knowledge retention [5, 2]. VR's ability to reconstruct physical environments has been widely adopted in museums [8, 6] to create experiences that captivate visitors and support education [9, 11].

Storytelling, an effective way to communicate complex ideas and engage audiences [1], has positively influenced attitudes toward science and improved comprehension in educational contexts [13]. Digital storytelling is defined as the combination of storytelling art with digital graphics, text, recorded audio narration, video, and music [3, 10]. The advent of VR offers a unique opportunity to digitize storytelling, enabling audiences to form emotional connections with exhibits and better understand historical [12] or scientific contexts [7]. This approach can help demystify complex topics like plant evolution in botanical education by enabling learners to interact with and visualize these processes engagingly. Building on these insights, our project merges VR and digital storytelling to create an interactive museum environment that explores the evolution and diversity of plants. Interactive features such as touch-triggered multimedia and a dynamic hand-held timeline guide users through the story of plant evolution. By merging narrative, multimedia, and spatial elements, the design is expected to deepen engagement, enhance knowledge retention, and motivate users to explore the diversity of plant life in meaningful ways.

## 3 CONCEPT AND IMPLEMENTATION

The design idea behind *Roots of Evolution* is to transform the Map of Plants (see Fig. 1) from a 2D visualisation into a 3D virtual museum, using the shape of the map as the layout of the rooms and corridors that the visitors can explore. The design preserves the map's colourful and approachable aesthetic, with warm tones and cartoon-like textures that ensure accessibility for all age groups while maintaining a cohesive visual style throughout the experience. Each section of the map is transformed into an interactive room in the museum representing an evolutionary branch of the plant kingdom. Visitors navigate through these rooms via doorways, each labeled with the corresponding plant group (e.g., Green Plants, Ferns, Monocots). Central to each room is a plant or group of plants represented by the drawings on the map, and visitors can interact with them through touch. Upon touch, multimedia content (videos, images, or animations) appears around the room (see

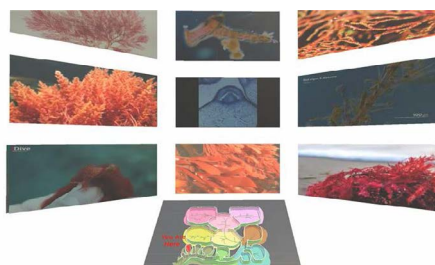


Figure 2: The video room with multiple videos, images, and a map showing the current location.

Fig. 2), giving real-life examples of the plants presented within their environment. There is also a map in front of visitors to show their current location, and they can go back to the museum by clicking the map. Each interaction (crossing a door or touching the plants) triggers a voiceover narration, offering a storytelling approach to explain the history of evolution and the significance of each plant.

They are also guided by a fun AI-driven virtual agent modeled after “Audrey II” a talking alien plant from the movie *Little Shop of Horrors* (1986). This plant can answer visitors’ questions about plants in the museum or on more general botany topics. Using the ConvAI<sup>2</sup> platform, we define its voice, backstory, personality, speaking style, and knowledge bank to ensure the information it provides is reliable. Some interactions are planned through narrative design. For example, if a user asks, “Where is the [plant name]?” the guide might respond: “Ah, you’re looking for [plant name], scientifically known as [scientific name]. It belongs to the [family name] family and is located in the [museum section].” With its friendly and knowledgeable personality, this virtual agent enhances visitors’ immersion, fosters curiosity, and makes learning about plants engaging and interactive.

The project is intended for a wide range of audiences, including students, families, and plant enthusiasts, and is suitable for learners of all ages, from middle school students to adults. The project aims to enhance understanding and cultivate curiosity about the natural world by presenting abstract concepts such as plant evolution through immersive storytelling. In addition to being used in museum exhibits, botanical gardens, and classroom settings, the VR experience is particularly suitable for informal learning environments where participants can explore at their own pace.

#### 4 FUTURE DEVELOPMENT

For future work, we aim to expand the VR museum to cover a more extensive range of topics in botany, such as the senses of plants, photosynthesis, reproduction, and other key biological processes. Using the same approach of digital storytelling, concepts from data physicalisation will be employed through interactive visualisations to make complex ideas more accessible. We will add more interactive visual representations. As visitors move through the museum, a timeline and a dynamic map appear on their hands in the VR environment, gradually highlighting the section explored. This map and timeline work as a dynamic exploration tool. As the visitors progress, they will unlock achievements for discovering specific plant groups and engaging content.

Additionally, we plan to further update the narrative design workflow to refine the virtual agent. Specifically, we will provide a navigation feature where a user asks where a relevant plant is located in the museum, and the virtual agent will take them to that room. We also plan to test the application with the public to assess its educational impact and engagement. We aim to work with

schools and science museums to reach diverse audiences while collaborating with teachers and experts to refine this prototype.

#### 5 CONCLUSION

In this paper, we present *Roots of Evolution*, an immersive VR museum that brings visitors through a journey of exploration of the evolution of plants. Through digital storytelling principles, *Roots of Evolution* makes complex topics like botany and evolution accessible and entertaining, helping people discover the vast diversity of the plant kingdom while fostering curiosity for further learning.

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<sup>2</sup><https://convai.com/>